Anthony Coache

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Degrees

Ph.D., Statistical Sciences (GPA: 3.95/4.00), University of Toronto — Oct 2024
- *Thesis: Risk-Sensitive Reinforcement Learning With Dynamic Risk Measures*

M.Sc., Statistics (GPA: 4.30/4.30), Université du Québec à Montréal — Jun 2020
- *Thesis: Stochastic Portfolio Optimization under Coherent Risk Measures*

B.Sc., Mathematics & Statistics (GPA: 4.18/4.30), Université du Québec à Montréal — Sep 2017

Work Experience

Imperial College London, Research Associate — Jun 2024 – Jun 2026
- *Led and collaborated on 4 concurrent projects in mathematical finance spanning risk modelling, statistical learning, optimisation, and market microstructure*
- *Supervised M.Sc. research projects conducted with leading banks and hedge funds, guiding model development and quantitative analysis*
- *Designed and taught courses in quantitative risk management and optimisation*

Oxford-Man Institute, University of Oxford, Visiting Researcher — Jan 2022 – Jun 2022
- *Developed a computationally efficient algorithm for risk-sensitive RL, leveraging elicitability to avoid nested simulations*
- *Invitation for a 6-month research visit from Prof. Álvaro Cartea*

Collaboration between Oanda and Fields-CQAM, Research Assistant — Jan 2020 – Sep 2020
- *Analyzed trends in order-flow trading data using hidden Markov models*
- *Backtested dynamic hedging strategies, improving Sharpe and tail-VaR*

University of Toronto, Ph.D. Candidate — Sep 2019 – Jul 2024
- *Published novel results in risk management and reinforcement learning with applications to dynamic decision-making under uncertainty*
- *Teaching assistant for 6 courses in statistics and financial insurance*

Université du Québec à Montréal, Research Intern — May 2016 – Aug 2017
- *Interdisciplinary works in statistics, portfolio optimization and epidemiology*
- *NSERC Undergraduate Student Research Awards & FRQNT Supplements*

Publications

A. Capponi, **A. Coache**, J. Muhle-Karbe. (2026) Optimal Trading in Automated Market Makers. **Submitted to INFORMS**.

M. Moghimi, **A. Coache**, H. Ku. (2026) Decoupling Time and Risk: Risk-Sensitive Reinforcement Learning with General Discounting. **Submitted to NeurIPS**.

Z. Cheng, **A. Coache**, S. Jaimungal. (2025) Eliciting Risk Aversion with Inverse Reinforcement Learning via Interactive Questioning. **Revised and resubmitted at JMLR**.

A. Coache, S. Jaimungal. (2025) Robust Reinforcement Learning with Dynamic Distortion Risk Measures. **SIAM J. Math. of Data Sci**.

A. Coache, S. Jaimungal, Á. Cartea. (2024) Conditionally Elicitable Dynamic Risk Measures for Deep Reinforcement Learning. **SIAM J. Financial Mathematics**.

A. Coache, S. Jaimungal. (2023) Reinforcement Learning with Dynamic Convex Risk Measures. **Mathematical Finance**.

Technical Skills

- *Programming: Python (PyTorch, Pandas, NumPy, matplotlib, sklearn, SciPy), TeX, Matlab, R, Git, SQL*
- *Interests: RL, risk measures, stochastic modeling, optimization, DeFi, applied statistics, music (piano)*
- *Languages: English, French*

Selected Scholarships & Awards

Mathematical Finance (Wiley) Top Cited Article of 2023 Publications — Mar 2025
NSERC Postdoctoral Fellowship — Sep 2024 – Aug 2026
UofT DoSS Doctoral Early Research Excellence Award — Jun 2023
SIAG/FME Conference Paper Prize — Jun 2023
Oxford-Man Institute Visitors Programme — Jan 2022
NSERC Alexander Graham Bell + **FRQNT** + **OGS** Doctoral Awards — Sep 2019 – Aug 2024
NSERC Alexander Graham Bell + **FRQNT** Master's Awards — Sep 2017 – Aug 2019

Selected Invited Talks

London-Paris Bachelier Workshop, London, United Kingdom — Nov 2025
SIAM Conference on Financial Mathematics and Engineering, Miami, USA — Jul 2025
ETH-HKG-ICL Mathematical Finance Workshop, Hong Kong, China — Apr 2025
King's College London Mathematical Finance Seminar, London, United Kingdom — Feb 2025
Control and Optimization Seminar at University of Connecticut, Online — Nov 2024
Mathematical Finance Seminar at Illinois Institute of Technology, Online — Oct 2024
Mathematical Insights from Markets, Control and Learning, Aussois, France — Sep 2024
STATQAM Seminar, Montréal, Canada — Feb 2024
SIAM Conference on Financial Mathematics and Engineering, Philadelphia, USA — Jun 2023
INFORMS Annual Meeting, Indianapolis, USA — Oct 2022
SIAM Conference on Financial Mathematics and Engineering, Online — Jun 2021

Teaching Experience

Imperial College London

- *Lecturer: Optimisation in Machine Learning* — Autumn 2025
- *Lecturer: Quantitative Risk Management* — Autumn 2024 & 2025
- *Lecturer: Convex Optimisation* — Autumn 2024

University of Toronto

- *Course instructor: Stochastic Methods for Actuarial Science* — Fall 2023
- *Workshop leader: Python and Matlab bootcamps for MFI program* — Fall 2021 & 2022
- *Teaching assistant: 6 courses in statistics and financial insurance* — Sep 2020 – Apr 2024

Université du Québec à Montréal

- *Teaching assistant: 4 courses in statistics* — Sep 2016 – Dec 2019

Student Supervision

M.Sc. Thesis at Imperial College London

- *Matthieu Fisher (Eisler Capital), Antoine Fuentes (BNP Paribas), Leo Goedhart, Zheyuan Li, Akram Zakriti (ExodusPoint)* — Summer 2025
- *Jeanne Gauthier (Morgan Stanley), Eloi Godier (Sakana Research), Linze Li (Ocean Leonid), Oussama Saadi (Deutsche Bank), Dean Yang (Qube)* — Summer 2024

Academic Service

Journal Referee

- *Quant. Finance (x3), SIAM J. Control and Optim. (x1), IMA J. Math. Control and Information (x1)*
- *NeurIPS Workshop Generative AI in Finance (x5), International Conference on AI in Finance (x17)*

Conference Organizing Committee

- *Co-organizer: Imperial College London's Finance & Stochastics Seminar Series* — Sep 2025 – May 2026
- *Translation committee and session chair: Canadian Statistics Student Conference* — Jun 2020 & 2021
- *Co-chair: First Edition of the Statistics Student Summit in Montréal* — Mar 2019

Github Contributions

- *Summarytools for Python, to generate comprehensive summaries within Jupyter* — Aug 2026

Last updated on August 13, 2026. Full academic CV available upon request.